

Lab Experiment-IV: The Bernoulli's apparatus

Objectives

- (i) Study the pressure variations in flow through a converging diverging nozzle.
- (ii) Compare the observations with Bernoulli's equation.
- (iii) Interpret the results and the potential discrepancies between theoretical analysis and observed data.

The Essential Background

1. Read up on Bernoulli's equation in Fluid Mechanics. Fundamentally, what is Bernoulli's equation about? What are the assumptions under which this equation is valid?
2. To circumvent the assumptions, several modifications to Bernoulli's equation have been proposed. Read up on these modifications from your Fluid Mechanics textbook and report them in your submission.
3. What is head loss in a pipe flow? Why is it important? Can you quantify head loss?
4. What is friction factor and what does it depend on?
5. How do you evaluate the friction factor for a given flow? Give examples.



Figure 4: Photo of the Bernuolli's apparatus.

Task 4.1: In the report for Lab Exp.-IV, concise answers to all the above questions have to be submitted.

About the experiment

Figure 4 shows a schematic of the experimental set up. A video of this set up as well as how to run the experiment has been uploaded on the google classroom. At the same time, a document with further details of the geometry of the set up has also been uploaded on google classroom. Watch the video very carefully to understand the set up and then carry out the experiment yourself and take all the required measurements. You may take help from the TAs and our Lab staff.

Task 4.2:

- (a) From the data that has been recorded by you, check whether the same satisfies the Bernoulli's equation. Also include the observed results in your report.
- (b) If not, state what factors might cause a mismatch between theory and experimental data.
- (c) Based on the “ Essential Background ” questions given at the beginning of this document, try to quantify as many of these “ factors ” as possible and subsequently incorporate them in the modified Bernoulli's equation (see Q2 in the Essential Background) in order to get a closer agreement between the theoretical prediction and the observations.
- (d) You have to clearly outline the procedure followed in performing the above tasks, including the numerical values of various quantities associated with the calculation.

Tasks 4.1 and 4.2 should be included in the report for this experiment.